



Database Management Systems

SQL overview, Data definition,
Structure of SQL queries

Anchor: Dr. Nagasundari S, Dept of CSE

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Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

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Dr. Nagasundari S

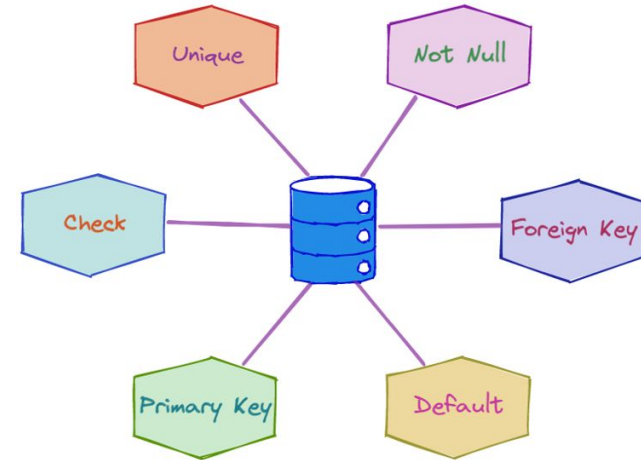
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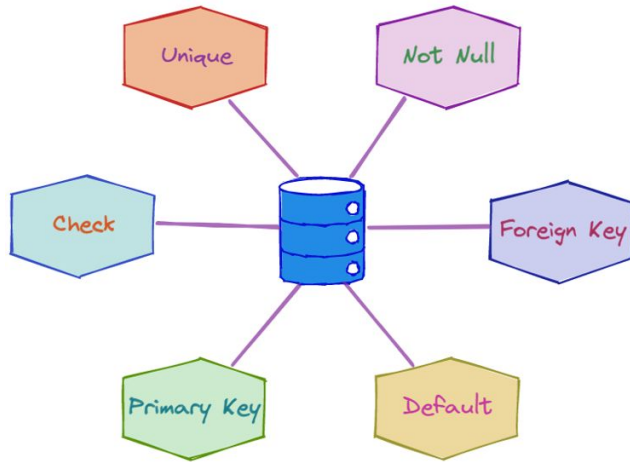
Specifying Constraints in SQL

- Constraints in SQL refer to the conditions and restrictions that are applied on the database
- This further means that before inserting data into the database, we are checking for some conditions.
- If the condition we have applied to the database holds true for the data that is to be inserted, then only the data will be inserted into the database tables.
- Constraints can be specified when the table is created with the CREATE TABLE statement, or after the table is created with the ALTER TABLE statement



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Specifying Constraints in SQL



Basic constraints:

Relational Model has 3 basic constraint types that are supported in SQL:

- **Key constraint:** A primary key value cannot be duplicated
- **Entity Integrity Constraint:** A primary key value cannot be null
- **Referential integrity constraints :** The “foreign key “ must have a value that is already present as a primary key, or may be null

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Specifying Constraints in SQL



Other Restrictions on attribute domains:

Default value of an attribute

- **DEFAULT** <value>
- NULL is not permitted for a particular attribute (NOT NULL)

CHECK clause

Dnumber INT NOT NULL CHECK (Dnumber > 0 AND Dnumber < 21);

NOT NULL

- Because SQL allows NULLs as attribute values, a constraint NOT NULL may be specified if NULL is not permitted for a particular attribute.
- This is always implicitly specified for the attributes that are part of the primary key of each relation, but it can be specified for any other attributes whose values are required not to be NULL

Example:

```
CREATE TABLE DEPARTMENT
( Dname VARCHAR(15) NOT NULL,
  Dnumber INT NOT NULL,
  Mgr_ssn CHAR(9) NOT NULL,
  Mgr_start_date DATE,
  PRIMARY KEY (Dnumber),
  UNIQUE (Dname),
  FOREIGN KEY (Mgr_ssn) REFERENCES EMPLOYEE(Ssn) );
```

DEFAULT

- It is also possible to define a default value for an attribute by appending the clause DEFAULT <value> to an attribute definition.
- The default value is included in any new tuple if an explicit value is not provided for that attribute
- **If no default clause is specified, the default default value is NULL** for attributes that do not have the NOT NULL constraint.

Example:

```
CREATE TABLE DEPARTMENT
(
    Dname VARCHAR(15) DEFAULT 'General',
    Dnumber INT NOT NULL DEFAULT 1,
    Mgr_ssn CHAR(9) NOT NULL,
    Mgr_start_date DATE,
    PRIMARY KEY (Dnumber),
    UNIQUE (Dname),
    FOREIGN KEY (Mgr_ssn) REFERENCES EMPLOYEE(Ssn)
);
```

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Specifying Constraints in SQL



Key Type	Enforces Uniqueness?	Allows NULLs?	Max Per Table	Main Purpose
Primary Key	Yes	No	Exactly 1	Row Identity
Foreign Key	No	Yes	Multiple	Table Relationships
Unique Key	Yes	Yes (Only one)	Multiple	Prevent Duplicate Data

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Specifying Constraints in SQL

Constraint Naming:

The CONSTRAINT keyword is followed by a constraint name followed by a column or a list of columns.

```
CREATE TABLE EMPLOYEE ( ... ,  
    Dno INT NOT NULL DEFAULT 1,  
    CONSTRAINT EMPPK PRIMARY KEY (Ssn),  
    CONSTRAINT EMPSUPERFK  
    FOREIGN KEY (Super_ssn) REFERENCES  
    EMPLOYEE(Ssn)  
    ON DELETE SET NULL ON UPDATE CASCADE,  
    CONSTRAINT EMPDEPTFK  
    FOREIGN KEY(Dno) REFERENCES  
    DEPARTMENT(Dnumber)  
    ON DELETE SET DEFAULT ON UPDATE CASCADE);
```

Types of SQL CONSTRAINTS

The SQL provides following types of CONSTRAINTS :

Constraint	Description
NOT NULL	This constraint confirms that a column cannot store NULL value.
UNIQUE	This constraint ensures that each row for a column must have a different value.
PRIMARY KEY	This constraint is a combination of a NOT NULL constraint and a UNIQUE constraint. This constraint ensures that the specific column or combination of two or more columns for a table have a unique identity which helps to find a particular record in a table more easily and quickly.
CHECK	A check constraint ensures that the value stored in a column meets a specific condition.
DEFAULT	This constraint provides a default value when specified none for this column.
FOREIGN KEY	A foreign key constraint is used to ensure the referential integrity of the data. in one table to match values in another table.

```
CREATE TABLE mytest(  
  agent_code char(6) NOT NULL UNIQUE ,  
  agent_name char(25) NOT NULL UNIQUE ,  
  working_area char(25) DEFAULT 'Mumbai',  
  commission decimal(8,2)  
  CHECK(commission>.1 AND commission<.3));
```

```
CREATE TABLE mytest(  
  agent_code char(6) NOT NULL UNIQUE ,  
  agent_name char(25) NOT NULL UNIQUE ,  
  working_area char(25) CHECK  
  ( working_area IN('London','Brisban','Chennai','Mumbai')) ,  
  commission decimal CHECK(commission<1));
```

```
CREATE TABLE DEPT_LOCATIONS ( Dnumber INT NOT NULL,  
Dlocation VARCHAR(15) NOT NULL DEFAULT 'Banglore',  
PRIMARY KEY (Dnumber, Dlocation),  
Constraint "fk_dno " FOREIGN KEY (Dnumber) REFERENCES  
DEPARTMENT(Dnumber) );
```

Referential actions

```
FOREIGN KEY (foreign_key_columns)  
REFERENCES parent_table(parent_key_columns)  
ON UPDATE action  
ON DELETE action;
```

FOREIGN KEY clause

Default operation: reject update on violation

- Attach **referential triggered action** clause
 - Options include SET NULL, CASCADE, and SET DEFAULT
 - Action taken by the DBMS for SET NULL or SET DEFAULT is the same for both ON DELETE and ON UPDATE
 - CASCADE option suitable for “relationship” relations

Delete actions of rows in the parent table

If you delete one or more rows in the parent table, you can set one of the following actions:

- **ON DELETE NO ACTION:** SQL Server raises an error and rolls back the delete action on the row in the parent table.
- **ON DELETE CASCADE:** SQL Server deletes the rows in the child table that is corresponding to the row deleted from the parent table.
- **ON DELETE SET NULL:** SQL Server sets the rows in the child table to NULL if the corresponding rows in the parent table are deleted. To execute this action, the foreign key columns must be nullable.
- **ON DELETE SET DEFAULT:** SQL Server sets the rows in the child table to their default values if the corresponding rows in the parent table are deleted. To execute this action, the foreign key columns must have default definitions. Note that a nullable column has a default value of NULL if no default value specified.

By default, SQL Server applies ON DELETE NO ACTION if you don't explicitly specify any action.

Update action of rows in the parent table

If you update one or more rows in the parent table, you can set one of the following actions:

- **ON UPDATE NO ACTION:** SQL Server raises an error and rolls back the update action on the row in the parent table.
- **ON UPDATE CASCADE:** SQL Server updates the corresponding rows in the child table when the rows in the parent table are updated.
- **ON UPDATE SET NULL:** SQL Server sets the rows in the child table to NULL when the corresponding row in the parent table is updated. Note that the foreign key columns must be nullable for this action to execute.
- **ON UPDATE SET DEFAULT:** SQL Server sets the default values for the rows in the child table that have the corresponding rows in the parent table updated.

- Additional Constraints on individual tuples within a relation are also possible using CHECK
- Apply to each tuple individually

```
CHECK (Dept_create_date <= Mgr_start_date);
```

Schema evolution commands

- DBA may want to change the schema while the database is operational
- Does not require recompilation of the database schema
- schema evolution commands available in SQL, which can be used to alter a schema by adding or dropping tables, attributes, constraints, and other schema elements.
- This can be done while the database is operational and does not require recompilation of the database schema checks must be done by the DBMS to ensure that the changes do not affect the rest of the database and make it inconsistent

DROP command

- The DROP command can be used to drop named schema elements, such as tables, domains, types, or constraints.
- One can also drop a whole schema if it is no longer needed by using the DROP SCHEMA command. There are two drop behavior options: CASCADE and RESTRICT.
- For example, to remove the COMPANY database schema and all its tables, domains, and other elements, the CASCADE option is used as follows

Drop behavior options:

CASCADE and RESTRICT

Example:

```
DROP SCHEMA COMPANY CASCADE;
```

This removes the schema and all its elements including tables, views, constraints, etc.

The DESCRIBE command in MySQL provides a detailed overview of a table's structure, including its column names, data types used, and any constraints applied to the columns.

```
DESCRIBE table_name;
```

(or)

```
SHOW COLUMNS FROM table_name;
```

(or)

```
DESC table_name;
```

SQL command **USE** is used to select a particular database created:

Syntax:

```
USE database_name;
```

To create table in the database the following query is used:

```
Create table table_name(.....)
```

To drop the database

```
DROP DATABASE [IF EXISTS] database_name;
```

Example:

```
Drop database customers;
```

1. Which command is used to delete all records from a table without removing its structure?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) ALTER

2. In SQL, what does the DESCRIBE command do?

- A) Shows the details of a table structure
- B) Deletes a table
- C) Modifies a table
- D) Creates a new schema

3. Which of the following statements is used to remove an entire table from the database?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) REMOVE

4. The ALTER TABLE statement is used to?

- A) Remove a table
- B) Modify an existing table structure
- C) Retrieve table data
- D) Define a new schema

Answers:

1. C
2. A
3. B
4. B



THANK YOU

Anchor: Dr. Nagasundari S, Dept of CSE

nagasundaris@pes.edu